

2	The quadratic function	Notes
	The manipulation of quadratic expressions. The roots of a quadratic equation. Simple examples involving functions of the roots of a quadratic equation.	Students should be able to factorise quadratic expressions and complete the square. Students should be able to use the discriminant to identify whether the roots are equal real, unequal real or not real. Students are expected to understand and use: $ax^2 + bx + c = 0$ has roots $\alpha, \beta = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ and forming an equation with given roots, which are expressed in terms of α and β: $\alpha + \beta = \frac{-b}{a} \text{ and } \alpha\beta = \frac{c}{a}$